CHANDU PRASAD BHAIRAPU



AI Developer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in internal applications and data workflows built with domain experts, from web applications, REST APIs and system integrations to automation, testing and cloud deployment on Azure and AWS. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by full-stack work in React and Python and close collaboration with chemists and domain experts across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Designed and optimized multi-agent document extraction with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) over complex regulatory and technical documents, raising precision/recall from 33%/45% to 93%/90% in evals against an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Sole developer owning the full knowledge-extraction pipeline end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Developed Expert Interface, an internal web application (React frontend, Python microservices backend) deployed on Azure, where domain experts validate and approve knowledge-graph changes.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph as a tool for agents, with millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Delivered as an AI-augmented engineer, using Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, React, FastAPI, Postgres, MongoDB, Neo4j, Docker, Kubernetes, Azure, LangGraph, LangChain, LiteLLM, MCP**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application with LangGraph that routes each request across vector database queries, general LLM inference and summarization, after mapping business stakeholders' decision-making workflows.
- Developed a Python backend with FastAPI and a React front-end, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, FastAPI, LangGraph, LangChain, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG generation and retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.

- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- **Tech stack: Python, LangChain, Hugging Face Transformers, PyTorch, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, CDK), Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Lab Requests Portal | TypeScript | React | PostgreSQL | Azure

- Internal web application where research teams file and track lab and equipment requests: React + TypeScript frontend, TypeScript (Node.js) REST API, PostgreSQL schema with migrations and row-level security, single sign-on with Entra ID and role-based access control, and a scheduled job that syncs equipment data.
- Automated tests (Vitest, Playwright), structured logging, request metrics and OpenTelemetry traces, CI/CD in GitHub Actions deploying to Azure Container Apps, and handover documentation for the long-term owner.

TECHNICAL SKILLSET

- **Software engineering:** Python, React (component-based frontend), FastAPI, REST APIs, Python microservices, Streamlit, Postman, API testing (Postman), debugging, clean code
- **Data:** SQL, PostgreSQL (relational databases), MongoDB, Neo4j, data pipelines, Pandas, Vector Databases (Qdrant, Pinecone, Azure AI Search)
- **Cloud & DevOps:** Azure, AWS (Lambda, SQS, DynamoDB, S3, EC2, CDK, serverless functions), Docker, Kubernetes, Git, GitLab CI/CD, infrastructure as code (AWS CDK)
- **AI & automation:** AI-assisted development (Claude Code), LangGraph, LangChain, LiteLLM, MCP, RAG, evaluation metrics (RAGAS, PytrecEval)
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI